RAGHAV PAI

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EDUCATION

College of Letters & Science, UC Berkeley, CA, USA 2026-Present
- *Freshman, Applied Mathematics Major*

Symbiosis International School, Pune, India 2018-2026
- *International Baccalaureate Diploma Programme (Grade 12)* 2024-2026
- *Cambridge IGCSE Examinations (Grade 10)* 2022-2024

STANDARDIZED TESTS

SAT: 1600 (Math: 800, English: 800) 2024

ACADEMIC HONORS AND AWARDS

- IB DP **44/45** 2024-2026
  - *7 in Math AA HL, Physics HL, Chemistry HL, Economics HL, English Lang & Lit SL, French Ab Initio SL 2 in EE/TOK)*
  - *Valedictorian*
- IGCSE **8 A*** 2022-2024
  - *World Topper (Computer Science)*
  - *National Topper (Physics, English)*
  - *ICE Distinction Certificate*

MATHEMATICS COMPETITIONS & HONORS

- University of Waterloo, CEMC contests 2024-2025
  (Distinction in Euclid; Honor Rolls in Hypatia, Fermat, Cayley, Galois)
- PROMYS India: Waitlisted 2025
- Indian Olympiad Qualifier in Mathematics: National & Regional Certificate 2024, 2025
- Regional Math Olympiad: Participated 2024
- AMC-12B: School Topper 2024

MATHEMATICS RESEARCH AND EXPERIENCES

Uncovering Periodicity in Pune's Climatic Cycles Using the Fourier Transform 2025
IB Extended Essay in Mathematics
Applied Fourier analysis to 75 years of rainfall data to extract dominant climatic cycles and relate them to climate change.

An Alternate Approach to Newcomb's Paradox through Probability and Game Theory 2025
Independent research submitted to MITx Philosophy Essay Award
Proposed a game-theoretic, probability-based resolution of Newcomb's paradox, challenging standard philosophical interpretations.

MITx's Paradox and Infinity (24.118x) 2025
Scored 97% in this undergraduate-level course on paradoxes, infinity, computability, and Gödel's Theorem; invited to submit to the MITx Philosophy Essay Award.

A Hyperbolic Geometry Approach to Epidemiological Dynamics 2024
Independent research under the guidance of Cheenta Academy
Modelled disease spread on hyperbolic networks, showing how non-Euclidean geometry better captures real-world epidemic dynamics. Paper published in the International Journal of Innovative Research In Technology (IJIRT), 12(5), 4096-4107.

SCHOOL LEADERSHIP

DP Prefect: Represented the DP cohort, coordinating student initiatives and bridging students and faculty. 2024-25

IG Lead: Led the IGCSE student body, fostering academic collaboration and organizing cohort-wide activities. 2023-24

Editor, *Sym'blem*: Directed the school magazine, curating student writing and managing publication. 2024-25

Literary Advisor: Guided student writers and led literary events, promoting a culture of creative expression 2025

EXTRACURRICULARS

Volunteer | Equal Community Foundation, Pune 2023-24

Supported gender-equality programs in low-income communities through workshops and content generation.

Volunteer | Inclusive Education Department, Symbiosis International School 2022-23, 2024-25

Led art-based cognitive sessions and ongoing mentoring for students with special needs.

Gold Prize | Poetry | Queen's Commonwealth Essay Competition 2023